The Gauge Structure Sub-Programme

Presentation Note 3

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Abstract

The gauge structure sub-programme identifies the internal symmetry groups that survive as invariants of admissible non-injective projection. Its central question is: which internal symmetry groups are forced by the admissibility structure of Π, rather than postulated? The sub-programme establishes the following chain:

$$\Pi^{-1}(O_n) \implies \text{phase fibre} \implies \mathrm{U}(1) \implies \mathrm{Im}\,\mathbb{H} \simeq \mathfrak{su}(2) \implies \mathrm{SU}(2) \implies \text{colour triplet} \implies \mathrm{SU}(3).$$

The status of each factor is distinct: U(1) is structural (Hopf fibre of $\Pi \simeq S^3$); SU(2) is closed analytically (O27–O30, Q6a); SU(3) is established unconditionally on the colour-adapted graph and on the standard graph at all four analytical levels (rank, averaged, effective-exponent, and pointwise), the last via Proposition 4.23 of O31 v1.5 [12] (single-frequency BI fingerprint structure). The sub-programme closes the gauge group identification. It does not derive Yang–Mills dynamics: that belongs to the gauge–gravity spectral stratification sub-programme.

Keywords. gauge structure; admissible projection; U(1); SU(2); SU(3); Hopf fibration; Weil representation; colour triplet; charge quantisation; gauge group; Yang–Mills; presentation note.

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1 Motivation and Central Question

The gauge structure sub-programme answers the following question.

Central question. The non-injective projection $\Pi : \chi \to O$ selects admissible configurations from the relational substrate. Its fibre $\Pi^{-1}(O_n)$ carries internal structure. *Which internal symmetry groups are preserved as invariants of admissible projection, and why are they precisely* U(1)*,* SU(2)*, and* SU(3)*?*

The question is stated precisely as follows. Define G_Π as the group of fibre symmetries of Π that preserve the admissibility constraints. The sub-programme establishes that $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

The physical importance of this identification is that it provides the gauge group of the Standard Model as a *forced output* of the admissibility structure, not as a postulate. In the Cosmochrony framework, gauge invariance is not introduced as a separate symmetry principle: it is the residual fibre symmetry of the projection. The gauge bosons and their interactions are subsequently derived as consequences of this structure via the spectral entropy functional (Q12, Q13).

This note is concerned exclusively with the *identification* of the gauge group G_Π. The derivation of Yang–Mills equations from the vertical variation of the spectral entropy belongs to the gauge–gravity spectral stratification sub-programme (Presentation Note 8) and is not claimed here.

2 Position in the Cosmochrony Programme

The gauge structure sub-programme sits at the intersection of Branch I (axiomatic primitive and fibre structure) and Branch II (spectral admissibility). It takes the following inputs from Presentation Notes 1 and 2 and uses them to identify the internal symmetry group.

From Note 1 (spectral admissibility):

- Spin-$\frac{1}{2}$ admissible sector $V_\rho \cong \mathbb{C}^2$ (O27–O29 [6, 2, 7]).
- Threshold dimension $\Sigma_c(n_3) = 3$ from quaternionic minimality (O23 [8]).
- $\mathrm{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$ as the spatial algebra (O23–O27).
- SU(3) colour sector unconditional on the standard graph via the single-frequency BI fingerprint mechanism of O31 v1.5 [12]; numerical confirmation in O32 [11].

From Note 2 (emergent geometry):

- Effective projection space $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$ (Q7 [13]).
- Effective base manifold $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ with metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Q5b–Q11).
- Admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$ with structure group G_Π (Q6a [4], Q12 [14]).

Remark 1 (Vocabulary note)**.** Throughout this note, *gauge invariance* refers to the residual fibre symmetry of the admissible projection Π, not to an independently postulated symmetry principle. The substrate χ is static; gauge freedom arises from the non-injectivity of Π, not from a dynamics of χ.

Remark 2 (Scope: gauge group identification only)**.** This sub-programme closes the identification $G_\Pi = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$. It does not derive the Yang–Mills equations or the gauge–gravity coupling ratio. Those results belong to Note 8 (Gauge–Gravity Stratification Sub-Programme).

3 Logical Chain of the Sub-Programme

The derivation chain has four stages.

$$\underbrace{\Pi \simeq S^3}_{\text{fibre topology}} \implies \underbrace{S^1 \hookrightarrow S^3 \to S^2}_{\text{Hopf fibration}} \implies \underbrace{\mathrm{U}(1)}_{\pi_1(S^1)\cong\mathbb{Z}} \implies \underbrace{\mathfrak{su}(2) \text{ from } \mathrm{Im}\,\mathbb{H}}_{\text{O23, O27}} \implies \underbrace{\mathrm{SU}(2)}_{\text{O27–O30, Q6a}} \implies \underbrace{\mathrm{SU}(3)}_{\text{O31–O32}}. \tag{1}$$

The three factors arise from distinct structural mechanisms.

1. U(1) *from the Hopf phase fibre.* The projection fibre $\Pi \simeq S^3$ carries a canonical Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$. The circle fibre S^1 supports a U(1) phase symmetry whose winding number $w \in \pi_1(S^1) \cong \mathbb{Z}$ is the topological origin of electric charge. This is a structural result: it follows from the minimal compact geometry required to support non-injective stable projections (TopologicalInvariants [9]).

2. SU(2) *from quaternionic admissibility.* The admissible sector is $V_\rho \cong \mathbb{C}^2$ (spin-$\frac{1}{2}$). The unique associative non-commutative minimal algebra compatible with admissibility is $\mathbb{H}$ (O23); $\dim_{\mathbb{R}} \mathrm{Im}\,\mathbb{H} = 3$ forces three internal directions. Any admissible morphism $\Phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$ (O27, quaternionic rigidity theorem). The identification chain is:

   $$\text{emergence} \Rightarrow \mathfrak{su}(2)\ [\text{O27}] \Rightarrow \text{pair data} \in \mathrm{Sym}(V_\rho, \mathbb{C})\ [\text{O29}] \Rightarrow r_{\text{eff}} = 3 \Rightarrow d_\rho = 2\ [\text{O29}].$$

3. SU(3) *from colour-triplet co-admissibility.* The metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph (O31). SU(3) is the symmetry group of co-admissible colour triplets $\{V_c, V_{\omega c}, V_{\omega^2 c}\}$ with $\omega^3 \equiv 1 \pmod q$, $q \equiv 1 \pmod 3$. On the standard graph, the identification holds at the rank, averaged, effective-exponent, and numerical levels.

4 Constituent Papers

4.1 The projective fibre and the U(1) phase sector

The U(1) factor arises from the topological structure of the projection fibre.

Foundation M [1] establishes that the projection fibre $\Pi^{-1}(O_n)$ is identified with the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, carrying a canonical symplectic (hence phase) structure.

TopologicalInvariants [9] classifies the admissible homotopy invariants of the projection fibre. The projection fibre $\Pi \simeq S^3$ carries the Hopf fibration $S^1 \hookrightarrow S^3 \xrightarrow{\pi_H} S^2$. Three results are established. (i) Electric charge arises from $\pi_1(S^1) \cong \mathbb{Z}$: the winding number $w \in \mathbb{Z}$ is the topological invariant, giving $q_{\text{eff}} = w \cdot e$. This is proved on the canonical admissible model (Proposition 3.1). (ii) On the canonical admissible model, the bounded-flux constraint removes the polar regions of the Hopf base S^2, reducing the admissible base to an annulus with $\pi_2(\mathcal{C}_{\text{adm}}) = 0$ (Proposition 4.1): no isolated magnetic monopoles are topologically admissible. (iii) Parity violation is interpreted as projective chirality: an orientation dependence of admissibility under Π, which selects one handedness of winding invariants.

Q6a [4] derives the gauge group G_Π, gauge charges, and the Yang–Mills principal bundle from the fibre structure of Π. The U(1) factor is identified as the phase sector of the Weil representation, arising from the canonical symplectic structure of V_ρ and its metaplectic lift. Q6a constructs the admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$ with structure group G_Π and connection $A \in \Omega^1(P, \mathfrak{g}_\Pi)$.

SMchargemass [3] uses the U(1) gauge-structure result as a downstream input, not as a constituent of its identification. Mass, charge, and inertia are interpreted there as saturated-response regimes of the same bounded projective mechanism. It is therefore an application of the gauge-structure output established here, and is listed under Outputs to Downstream Branches rather than as a constituent paper.

4.2 Quaternionic admissibility and the SU(2) sector

The SU(2) factor is closed analytically by the chain O23–O30 and Q6a.

O23 [8] proves that the threshold $\Sigma_c(n_3) = 3$ follows from the fact that $\mathbb{H}$ is the minimal admissible associative non-commutative algebra, and $\dim_{\mathbb{R}} \operatorname{Im} \mathbb{H} = 3$. Three is not an input; it is derived from the quaternionic structure.

O27 [6] proves the quaternionic rigidity theorem: every admissible morphism $\Phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$. This is the analytical cornerstone of the SU(2) identification. It is proved unconditionally (Theorem 1.1 of O27), without reference to the bridge conjecture or the finite-q structure.

O28 [2] establishes asymptotic calibration: the covariance $\mathcal{C}_c$ in $\mathrm{End}(H_{\mathrm{eff}})$ has rank exactly 3 for every conjugate pair at $q \in \{61, 101, 151\}$, with invariant eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$ (the exact value $\lambda_1/\lambda_2 = 2$ is derived analytically in O30).

O29 [7] identifies the spin-$\frac{1}{2}$ irreducible sector via the symmetric rank formula $r_{\mathrm{eff}} = d_\rho(d_\rho + 1)/2$. The unique solution $d_\rho = 2$ is confirmed for $q \in \{61, 101, 151\}$.

O30 [10] derives analytically the invariant eigenvalue ratio $\lambda_1/\lambda_2 = 2$: Born–Infeld parity anti-linearity forces an equatorial constraint on the admissible fibre. The factor of two is purely structural (squared symmetric-tensor normalisation of the central Carnot generator $Z = [X, Y]$).

Q6a [4] closes the SU(2) identification at the level of the admissible principal bundle: the SU(2) factor is the structure group of the non-abelian fibre sector corresponding to the spin-$\frac{1}{2}$ module. The associated connection is the admissible gauge field.

Q7 [13] establishes that the three-dimensional internal space $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$ carries the unique irreducible $\mathfrak{su}(2)$-module of dimension 3, confirming that the spatial and internal $\mathfrak{su}(2)$ structures are representation-theoretically locked.

4.3 Gauge charges and admissible bundle structure

Q6a [4] constructs the full admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$ with fibre structure matching the three admissibility fixed points of orbit sizes 1, 2, and 3. Gauge charges are identified as holonomy invariants of the admissible connection A. The derivation is structural and does not require the Yang–Mills equations.

Q12 [14] identifies $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ explicitly as the composite of orbit sub-bundles and extends A_g to the Laplace-type operator $A_{g,A}$ on the associated vector bundle. The base–fibre decomposition and the horizontal–vertical decoupling lemma (Lemma 1 of Q12) guarantee that the gauge sector a_4 and the gravitational sector a_2 are independent.

TopologicalInvariants [9] places the charge identification within the topological layer: charge is a π_1 invariant; the absence of monopoles follows from $\pi_2(\mathcal{C}_{\mathrm{adm}}) = 0$ on the canonical model.

4.4 Colour-triplet co-admissibility and the SU(3) sector

The SU(3) sector arises from the co-admissibility of Weil-block triplets under the $\mathbb{Z}/3\mathbb{Z}$ symmetry of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for $q \equiv 1 \pmod 3$.

O31 [12] develops the full SU(3) emergence framework. Three results are established. (i) Metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for

the colour-adapted Cayley graph $\mathrm{Cay}(G_q, S_q^{(3)})$ (proved). (ii) SU(3) is identified as the symmetry group of the co-admissible colour triplet $\{V_c, V_{\omega c}, V_{\omega^2 c}\}$, proved for the standard graph via [H-color] (Proposition 4.23 of O31, proved analytically). (iii) The spectral route to [H-color] is closed: exact generator-twist isospectrality does not imply colour-triplet capacity profile equality (proved, closing the spectral approach).

O32 [11] provides the numerical test and the partial analytical resolution of [H-color]. Four levels are established. $[\text{H-color}]_{\text{rank}}$: sector rank equality $r_c(n) = r_{\omega c}(n) = r_{\omega^2 c}(n)$ proved unconditionally via intra-coset Vandermonde symmetry (O31). $[\text{H-color}]_{\text{avg}}$: block-averaged capacity equality $\mathbb{E}[\sigma_c(n)] = \mathbb{E}[\sigma_{\omega c}(n)] + O(q^{-1})$, proved in O32 via the modulation frequency argument. $[\text{H-color}]_{\text{eff}}$: effective exponent equality $\delta_c = \delta_{\omega c} = \delta_{\omega^2 c}$ in the $q \to \infty$ limit, proved in O32. $[\text{H-color}]_{\text{pointwise}}$: exact profile equality at finite q, proved analytically in O31 (Proposition 4.23) via the single-frequency fingerprint structure: each O12 fingerprint vector carries frequency $B = c_1 b_1 + c_2 b_2 + c_3 b_3$, and the orbit dilation $B \mapsto \omega B$ is a bijection of $\mathbb{Z}/q\mathbb{Z}$. Numerical results: passes for $q \in \{61, 151, 211, 307\}$ ($q \equiv 1 \pmod 3$, test primes); controls $q \in \{29, 101\}$ ($q \equiv 2 \pmod 3$) confirm noise floor.

4.5 Boundary with Yang–Mills dynamics

The gauge structure sub-programme identifies G_Π and constructs the admissible principal bundle. It does not derive the Yang–Mills equations. That derivation belongs to the gauge–gravity spectral stratification sub-programme (Presentation Note 8), where the vertical variation of the joint spectral entropy functional $\mathcal{S}_\Pi[g, A]$ yields the a_4 Yang–Mills response.

The logical boundary is sharp: Q6a and Q12 are constituent papers of the gauge structure sub-programme in so far as they identify the gauge group and construct the bundle. Q12 and Q13 become constituent papers of Note 8 in so far as they derive the Yang–Mills equations and the gauge–gravity coupling.

The a_4-level result of Q12 (Theorem 1) depends on the gauge group G_Π as an input from Q6a and the O-series. It depends on [H-color] for the SU(3) sector in exactly the same way as the gauge group identification, with [H-color] now proved by Proposition 4.23 of O31 [12].

Table 1: Summary of constituent papers by sector. Status: P = proved; S = structural; N = numerical; O = open.

Sector	Papers	Central output	Status
U(1) fibre	Foundation, TopInv	Charge $= w \cdot e$; no monopoles	P (canonical)
U(1) bundle	Q6a, Q12	Admissible U(1) connection	S
SU(2) sector	O23, O27–O30, Q6a, Q7	$d_\rho = 2$; $\mathfrak{su}(2)$ closed	P
Charges	Q6a, TopInv, Q12	Bundle structure; holonomy charges	S
SU(3) adapted	O31	Triplet co-admissibility	P
SU(3) standard	O31–O32	$[\text{H-color}]_{\text{rank/avg/eff/pointwise}}$	P
$[\text{H-color}]_{\text{pointwise}}$	O31	Proved analytically (Prop. 4.23)	P

5 Inputs from Upstream Results

1. **Weil representation and fibre structure.** The identification $\Pi^{-1}(O_n) \cong V_\rho$ is a theorem of Foundation M [1]. The canonical symplectic structure of V_ρ is the origin of the phase degree of freedom.

2. **Non-injectivity as structural necessity.** ENI [5] establishes that non-injectivity is not

an interpretive option: any genuinely emergent description with $S_\Pi > 0$ must be non-injective. Gauge freedom is a direct consequence of this non-injectivity: the fibre $\Pi^{-1}(O_n)$ is non-trivial precisely because Π is non-injective.

3. **Spin-$\frac{1}{2}$ sector and $\Sigma_c(n_3) = 3$.** From Note 1 (O23, O27, O29): the admissible sector is $V_\rho \cong \mathbb{C}^2$; $\mathrm{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$ carries three internal directions; $d_\rho = 2$ is the unique admissible irreducible dimension.

4. **Effective geometry.** From Note 2 (Q5b–Q11): the effective base manifold $M = \mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ is the space over which the principal bundle is defined.

6 Outputs to Downstream Branches

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
$G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$	Q12, Q13, Q14	P (O31 v1.5)
Charge/mass/inertia as saturated responses	SMchargemass [3]	S
Admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$	Q12, Q13	S
Charge quantisation $q_{\text{eff}} = w \cdot e$	Q14, applications	P (canonical)
Absence of magnetic monopoles	Q14	P (canonical)
$d_\rho = 2$, $\mathfrak{su}(2)$ closure	Q14, Q12, Q13	P
SU(3) at $[\text{H-color}]_{\text{pointwise}}$ level	Q12, Q13, O31	P (exact at finite q)
Projective chirality	Q14	S

7 Status of Results

7.1 Proved results

1. *Charge quantisation on the canonical model* (TopologicalInvariants [9], Proposition 3.1): $q_{\text{eff}} = w \cdot e$ with $w \in \mathbb{Z} = \pi_1(S^1)$.

2. *Absence of magnetic monopoles on the canonical model* (TopologicalInvariants, Proposition 4.1 and Corollary 5.1): $\pi_2(\mathcal{C}_{\text{adm}}) = 0$ on the canonical admissible model; no isolated magnetic charge is topologically admissible.

3. *Quaternionic minimality* (O23 [8]): $\mathbb{H}$ is the minimal admissible associative non-commutative algebra; $\dim_{\mathbb{R}} \mathrm{Im}\,\mathbb{H} = 3$; three is derived, not postulated.

4. *Quaternionic rigidity* (O27 [6], Theorem 1.1): every admissible morphism $\Phi_{q,\rho}$ factors through $\mathfrak{su}(2)$, proved unconditionally.

5. *Spin-$\frac{1}{2}$ sector identification* (O29 [7]): $r_{\text{eff}} = 3$, $d_\rho = 2$, confirmed for $q \in \{61, 101, 151\}$.

6. *Eigenvalue ratio* $\lambda_1/\lambda_2 = 2$ (O30 [10]): analytically derived from the equatorial fibre constraint.

7. *Triplet co-admissibility on the colour-adapted graph* (O31 [12]): metaplectic intertwining forces unconditional co-admissibility for $\mathrm{Cay}(G_q, S_q^{(3)})$.

8. *Generator-twist isospectrality* (O31, proved): the spectral route to [H-color] is closed.

9. $[\text{H-color}]_{\text{rank}}$ (O31): intra-coset Vandermonde symmetry gives rank equality unconditionally.

10. $[\text{H-color}]_{\text{avg}}$ (O32 [11], Proposition 1): modulation frequency argument gives $\mathbb{E}[\sigma_c(n)] = \mathbb{E}[\sigma_{\omega c}(n)] + O(q^{-1})$.

11. $[\text{H-color}]_{\text{eff}}$ (O32): effective exponent equality $\delta_c = \delta_{\omega c} = \delta_{\omega^2 c}$ in the $q \to \infty$ limit.

12. *Horizontal–vertical decoupling* (Q12 [14], Lemma 1): the Einstein sector a_2 and the Yang–Mills sector a_4 are independent under the joint spectral entropy; the gauge group structure does not affect the gravitational response at leading order.

7.2 Structural results

1. U(1) *as phase fibre* (Foundation, Q6a): the U(1) factor is the phase symmetry of the canonical symplectic structure of V_ρ. The identification is structural; it follows from the Weil representation and is not a separate postulate.

2. SU(2) *bundle sector* (Q6a): the SU(2) factor is the structure group of the non-abelian fibre sector; the admissible gauge field is the associated connection.

3. SU(3) *on the standard graph* (O31 v1.5, Proposition 4.23, unconditional via single-frequency BI fingerprint): SU(3) is the symmetry group of the co-admissible colour triplet for $q \equiv 1 \pmod 3$. The identification $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ holds unconditionally on the standard graph and is structural at the orbit-size level (orbit sizes 3, 2, 1 respectively).

4. *Projective chirality and parity* (TopologicalInvariants, §6): parity violation arises from the orientation dependence of admissibility under Π; the $V - A$ structure is identified as the maximal chirality class (partial structural result).

5. *Gauge charges as holonomy invariants* (Q6a): the admissible connection A and its curvature $F_{\mu\nu}$ encode gauge charges as holonomy classes of the bundle.

7.3 Numerical evidence

1. $r_{\text{eff}} = 3$ *and* $d_\rho = 2$ (O28–O29): confirmed at $q \in \{61, 101, 151\}$ with zero inter-pair variance for $q \in \{61, 151\}$.

2. $[\text{H-color}]_{\text{pointwise}}$ (O32): numerical passage for $q \in \{61, 151, 211, 307\}$ with variance ratio $R_{\text{var}}(q)$ decreasing toward zero.

7.4 Open hypotheses

1. $[\text{H-color}]_{\text{pointwise}}$: proved analytically in O31 (Proposition 4.23) via the single-frequency structure of the O12 fingerprint: each vector carries frequency $B = c_1 b_1 + c_2 b_2 + c_3 b_3$, and the orbit dilation $B \mapsto \omega B$ is a bijection of $\mathbb{Z}/q\mathbb{Z}$, so $\sigma_c(n) = \sigma_{\omega c}(n) = \sigma_{\omega^2 c}(n)$ holds exactly. The SU(3) identification on the standard graph is therefore fully unconditional.

2. $\pi_2(C_{\text{eff}}) = 0$ *on the full configuration space* (TopologicalInvariants, Remark 4.2): the monopole exclusion is proved on the canonical model. Extension to the full effective configuration space C_{eff} is open.

3. *Maximal chirality ($V - A$)* (TopologicalInvariants, Remark 6.2): projective chirality is established as a structural mechanism. Its identification with the specific $V - A$ structure of the weak interaction remains qualitative.

8 Open Deliverables

Three open items define the boundary of the sub-programme.

Closed: $[\text{H-color}]_{\text{pointwise}}$**.** All four levels of [H-color] are now established analytically. The SU(3) identification on the standard graph is fully unconditional: the single-frequency structure of the O12 fingerprint (Lemma 4.22 of O31) gives $\sigma_c(n) = \sigma_{\omega c}(n) = \sigma_{\omega^2 c}(n)$ exactly at finite q (Proposition 4.23 of O31 [12]), completing the identification of G_Π without any residual conditionality.

Open deliverable 2: $\pi_2(C_{\text{eff}}) = 0$ **in general.** The topological exclusion of magnetic monopoles is proved on the canonical admissible model. Extension to the full C_{eff} is a well-posed open problem in homotopy theory.

Open deliverable 3: $V - A$ **maximality from admissibility.** The structural mechanism for projective chirality is identified. Its derivation of the specific $V - A$ form of the weak interaction current from admissibility data is an open analytical problem.

9 Conclusion

The gauge structure sub-programme identifies the admissible internal symmetry group $G_\Pi = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ as a forced output of the admissibility constraints on the projection fibre Π.

The three factors have distinct origins and distinct analytical status. U(1) arises from the Hopf phase fibre of $\Pi \simeq S^3$; charge quantisation follows from $\pi_1(S^1) \cong \mathbb{Z}$. SU(2) arises from the quaternionic minimality of the admissible algebra and the rigidity theorem of O27; it is closed analytically. SU(3) arises from the metaplectic co-admissibility of Weil-block colour triplets; it is established unconditionally on the colour-adapted graph and at all four analytical levels ($[\text{H-color}]_{\text{rank}}$, $[\text{H-color}]_{\text{avg}}$, $[\text{H-color}]_{\text{eff}}$, $[\text{H-color}]_{\text{pointwise}}$) on the standard graph, the last via Proposition 4.23 of O31 [12]. The sub-programme closes the gauge group identification without residual conditionality.

This sub-programme closes the gauge group identification. It does not derive Yang–Mills dynamics: the a_4 Yang–Mills response from the vertical variation of the spectral entropy belongs to Note 8 (Gauge–Gravity Stratification).

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